# Anirudh Phukan

+91-9902542290 anirudh.phukan@gmail.com nKfP2coAAAAJ anirudh-phukan.github.io

## Research Interests

Continual Learning · Reasoning · Agentic RAG · Interpretability · Factuality & Attribution

## Education

**Ph.D. Indian Institute of Science** *July 2025 – Present*
*Department of Computational and Data Sciences*
*Advisor: Prof. Danish Pruthi*
CGPA: 9.30/10.00

**Bachelor of Technology, Indian Institute of Technology Guwahati** *July 2019 – May 2023*
*Major in Computer Science and Engineering with Minor in Robotics and AI*
CGPA: 9.59/10.00

## Experience

**Research Associate at Adobe Research, India** *July 2023 – July 2025*

- Contributed to projects on Document Grounding, Hallucination Detection and Mitigation, LLM Optimization, Multilinguality, and Multimodality
- Facilitated cross-functional and global research collaboration, strategically engaging with diverse stakeholders to align cutting-edge research with strategic business objectives
- Published **4 papers** and filed **6 patents** in my time here. Successfully **productized** entity attribution in contracts in Adobe Acrobat. Highlighted by Adobe's CEO in a *Fox Business interview* (6:45–7:15)

**Research Intern at Adobe Research, India** *May 2022 – July 2022*

- Developed and **patented** a framework to deconstruct infographics into components and reformat them for improved readability across various screen sizes while preserving the original reading order

## Preprints

1. **Anirudh Phukan**, Tanya Goyal, and Danish Pruthi. **BlindToChange: AI Agents Can Find Changes When Asked But Do They Look When Needed?**, *arXiv preprint arXiv:2610.41435, 2026.*
2. Nikhil Anand, Shwetha Somasundaram, **Anirudh Phukan**, Apoorv Saxena, and Koyel Mukherjee. **ContextFocus: Activation Steering for Contextual Faithfulness in Large Language Models**, *arXiv preprint arXiv:2601.04131, 2026.*

## Publications

1. Agam Goyal, Koyel Mukherjee, Apoorv Saxena, **Anirudh Phukan**, Eshwar Chandrasekharan, and Hari Sundaram. **Masking or Mitigating? Deconstructing the Impact of Query Rewriting on Retriever Biases in RAG**, *In Findings of the Association for Computational Linguistics: ACL 2026.*
2. **Anirudh Phukan**, Divyansh, Harshit Kumar Morj, Vaishnavi, Apoorv Saxena, and Koustava Goswami. **Beyond Logit Lens: Contextual Embeddings for Robust Hallucination Detection & Grounding in VLMs**, *In Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies.*
3. Shwetha Somasundaram, **Anirudh Phukan**, and Apoorv Saxena. **PLD+: Accelerating LLM inference by leveraging Language Model Artifacts**, *In Findings of the Association for Computational Linguistics: NAACL 2025.*
4. **Anirudh Phukan**, Shwetha Somasundaram, Apoorv Saxena, Koustava Goswami, and Balaji Vasan Srinivasan. **Peering into the Mind of Language Models: An Approach for Attribution in Contextual Question Answering**, *In Findings of the Association for Computational Linguistics: ACL 2024.*

5. S. Kumar*, **A. Phukan*** and A. Sur. **IPCL: Iterative Pseudo-Supervised Contrastive Learning to Improve Self-Supervised Feature Representation**, *ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP).*

## Patents

1. Nikhil Anand, **Anirudh Phukan**, Shwetha S, Apoorv Saxena, Koyel Mukherjee. **Context-Focused Steering for Machine-Learning Models** [Filed] (US Patent App. 19/229,734)
2. **Anirudh Phukan**, Animesh Kumar, Apoorv Saxena, Koustava Goswami. **Attribution in Contextual Question Answering using Machine-Learning Hidden States** [Filed] (US Patent App. 19/065,659)
3. **Anirudh Phukan**, Koustava Goswami, Divyansh, Harshit Kumar Morj, Vaishnavi. **Generating Multimodal Attribution of Artificial Intelligence Responses** [Filed] (US Patent App. 18/927,104)
4. Shwetha S., **Anirudh Phukan**, Apoorv Saxena. **Generating Draft Sequence rankings for Speculative Decoding using Large Language Model Hidden States** (US Patent No. 12,639,354)
5. Inderjeet Nair, **Anirudh Phukan**, Aravind Veluri, Lakshya J., Mohar Kundu, Akhash Amarnath, Niyati Chhaya, Sumit Shekhar. **Reflowing Infographics for Enhanced Cross-Device Consumption** (US Patent No. 12,443,790)

## Awards and Honors

**Google PhD Fellowship in Natural Language Processing** *2026*

**Undergraduate Distinctions** *2019 - 2023*

- Achieved $3^{rd}$ Rank in Computer Science and Engineering among 111 Students, Class of 2023
- Awarded Outstanding Grade AS in 5 courses including *Deep Learning* and *Optimization*

**Engineering Entrance Examinations** *2019*

- Secured All India Rank 1,974 in JEE Advanced among 250,000 shortlisted candidates (99.2 Percentile) and All India Rank 1,201 in JEE Mains among 1,500,000 applicants (Top 0.1%)
- Secured $10^{th}$ rank in KCET among $\sim$ 200,000 applicants

**Kishore Vaigyanik Protsahan Yojana (KVPY) Scholar** *2019*

- Awarded by Dept. of Science and Technology, Govt. of India to students with an aptitude for research
- All India Rank 404 among $\sim$ 100,000 applicants (Top 0.5%)

**National Talent Search (NTSE) Scholar** *2017*

- Conducted by NCERT to identify and nurture talented students
- Acceptance rate of $< 0.1\%$ out of $\sim$ 1,000,000 applicants

**Standardized Tests**

- GRE - 334/340, TOEFL - 114/120

## Service, Mentoring & Misc.

- **Reviewer:** ICLR 2027; ACL Rolling Review (ARR): Oct 2024, Dec 2024, Feb 2025, May 2025, Jan 2026, May 2026
- Co-mentored 7 Undergraduate students in **Adobe Research India**'s Summer Internship program (2024-2025)
- CSEA Department Academic Mentor to 5 freshmen, helping them cope up with the curriculum and solving their general concerns.